FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: buried
Model status: BOUNDED_IEC60287_SINGLE_CORE_HOMOGENEOUS_DIRECT_BURIAL
Captured UTC: 2026-10-26T06:04:35.663Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: installation | Entered value: single-core-homogeneous-direct-burial
Input path: conductorDiameterMM | Entered value: 20
Input path: insulationLayers[0].innerDiameterMM | Entered value: 20
Input path: insulationLayers[0].outerDiameterMM | Entered value: 31
Input path: insulationLayers[0].thermalResistivityKmPerW | Entered value: 3.5
Input path: beddingLayers[0].innerDiameterMM | Entered value: 31
Input path: beddingLayers[0].outerDiameterMM | Entered value: 34
Input path: beddingLayers[0].thermalResistivityKmPerW | Entered value: 4.5
Input path: jacketLayers[0].innerDiameterMM | Entered value: 34
Input path: jacketLayers[0].outerDiameterMM | Entered value: 43
Input path: jacketLayers[0].thermalResistivityKmPerW | Entered value: 3.5
Input path: burialDepthToAxisMM | Entered value: 900
Input path: soilThermalResistivityKmPerW | Entered value: 1
Input path: ambientSoilC | Entered value: 25
Input path: conductorLimitC | Entered value: 90
Input path: dielectricLossWPerM | Entered value: 0.05
Input path: sheathLossFactor | Entered value: 0.1
Input path: armourLossFactor | Entered value: 0
Input path: centerOffsetsMM[0] | Entered value: -200
Input path: centerOffsetsMM[1] | Entered value: 0
Input path: centerOffsetsMM[2] | Entered value: 200
Input path: acResistanceAtLimitOhmPerM | Entered value: 0.00013
Input path: acResistanceSource | Entered value: ILLUSTRATIVE INPUT ONLY: replace with a manufacturer-qualified AC conductor resistance at rated conductor temperature
Input path: loadCurrentA | Entered value: 250

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_IEC60287_SINGLE_CORE_HOMOGENEOUS_DIRECT_BURIAL
Calculated output path: installation | Value: single-core-homogeneous-direct-burial
Calculated output path: scope | Value: One or a flat row of equally loaded identical single-core cables in uniform, undried soil at equal burial depth. Not trefoil, duct banks, cyclic/drying-out or heterogeneous soil.
Calculated output path: cableOuterDiameterMM | Value: 43
Calculated output path: burialDepthToAxisMM | Value: 900
Calculated output path: conductorResistanceOhmPerM | Value: 0.00013
Calculated output path: acResistanceSource | Value: ILLUSTRATIVE INPUT ONLY: replace with a manufacturer-qualified AC conductor resistance at rated conductor temperature
Calculated output path: t1KmPerW | Value: 0.24412653507231694
Calculated output path: t2KmPerW | Value: 0.06615751728897509
Calculated output path: t3KmPerW | Value: 0.13081558613776706
Calculated output path: isolatedExternalThermalResistanceKmPerW | Value: 0.7046340497704088
Calculated output path: cableRatings[0].index | Value: 0
Calculated output path: cableRatings[0].centerOffsetMM | Value: -200
Calculated output path: cableRatings[0].mutualThermalResistanceKmPerW | Value: 0.5938927293857521

Calculated output path: cableRatings[0].externalThermalResistanceKmPerW | Value: 1.2985267791561608
Calculated output path: cableRatings[0].continuousRatingA | Value: 514.1364394513777
Calculated output path: cableRatings[0].dielectricTemperatureRiseC | Value: 0.08087815750595306
Calculated output path: cableRatings[0].heatingCoefficient | Value: 0.0002455929327687563
Calculated output path: cableRatings[1].index | Value: 1
Calculated output path: cableRatings[1].centerOffsetMM | Value: 0
Calculated output path: cableRatings[1].mutualThermalResistanceKmPerW | Value: 0.7013511510203021
Calculated output path: cableRatings[1].externalThermalResistanceKmPerW | Value: 1.405985200790711
Calculated output path: cableRatings[1].continuousRatingA | Value: 498.748727522147
Calculated output path: cableRatings[1].dielectricTemperatureRiseC | Value: 0.08625107858768058
Calculated output path: cableRatings[1].heatingCoefficient | Value: 0.000260959487062497
Calculated output path: cableRatings[2].index | Value: 2
Calculated output path: cableRatings[2].centerOffsetMM | Value: 200
Calculated output path: cableRatings[2].mutualThermalResistanceKmPerW | Value: 0.5938927293857521
Calculated output path: cableRatings[2].externalThermalResistanceKmPerW | Value: 1.2985267791561608
Calculated output path: cableRatings[2].continuousRatingA | Value: 514.1364394513777
Calculated output path: cableRatings[2].dielectricTemperatureRiseC | Value: 0.08087815750595306
Calculated output path: cableRatings[2].heatingCoefficient | Value: 0.0002455929327687563
Calculated output path: limitingCableIndex | Value: 1
Calculated output path: groupContinuousCurrentRatingA | Value: 498.748727522147
Calculated output path: enteredLoadCurrentA | Value: 250
Calculated output path: withinGroupRating | Value: true
Calculated output path: warnings[0] | Value: AC resistance skin/proximity correction inputs, dielectric losses and sheath/armour loss factors require manufacturer qualification.
Calculated output path: warnings[1] | Value: This is a constrained homogeneous direct-burial steady-state calculation and is NOT a verified full IEC 60287 installation/NEC ampacity rating.
Calculated output path: warnings[2] | Value: Soil dry-out, thermal backfill, ducts, other heated circuits and cyclic/seasonal conditions are excluded.
Calculated output path: warnings[3] | Value: Use project-specific thermophysical and conductor data and engineering review for final cable sizing.

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: buried
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:35.663Z
Evidence / limitation: outputs.scope | Value: One or a flat row of equally loaded identical single-core cables in uniform, undried soil at equal burial depth. Not trefoil, duct banks, cyclic/drying-out or heterogeneous soil.
Evidence / limitation: outputs.warnings[0] | Value: AC resistance skin/proximity correction inputs, dielectric losses and sheath/armour loss factors require manufacturer qualification.
Evidence / limitation: outputs.warnings[1] | Value: This is a constrained homogeneous direct-burial steady-state calculation and is NOT a verified full IEC 60287 installation/NEC ampacity rating.
Evidence / limitation: outputs.warnings[2] | Value: Soil dry-out, thermal backfill, ducts, other heated circuits and cyclic/seasonal conditions are excluded.
Evidence / limitation: outputs.warnings[3] | Value: Use project-specific thermophysical and conductor data and engineering review for final cable sizing.